Multilevel Modeling for Longitudinal Data

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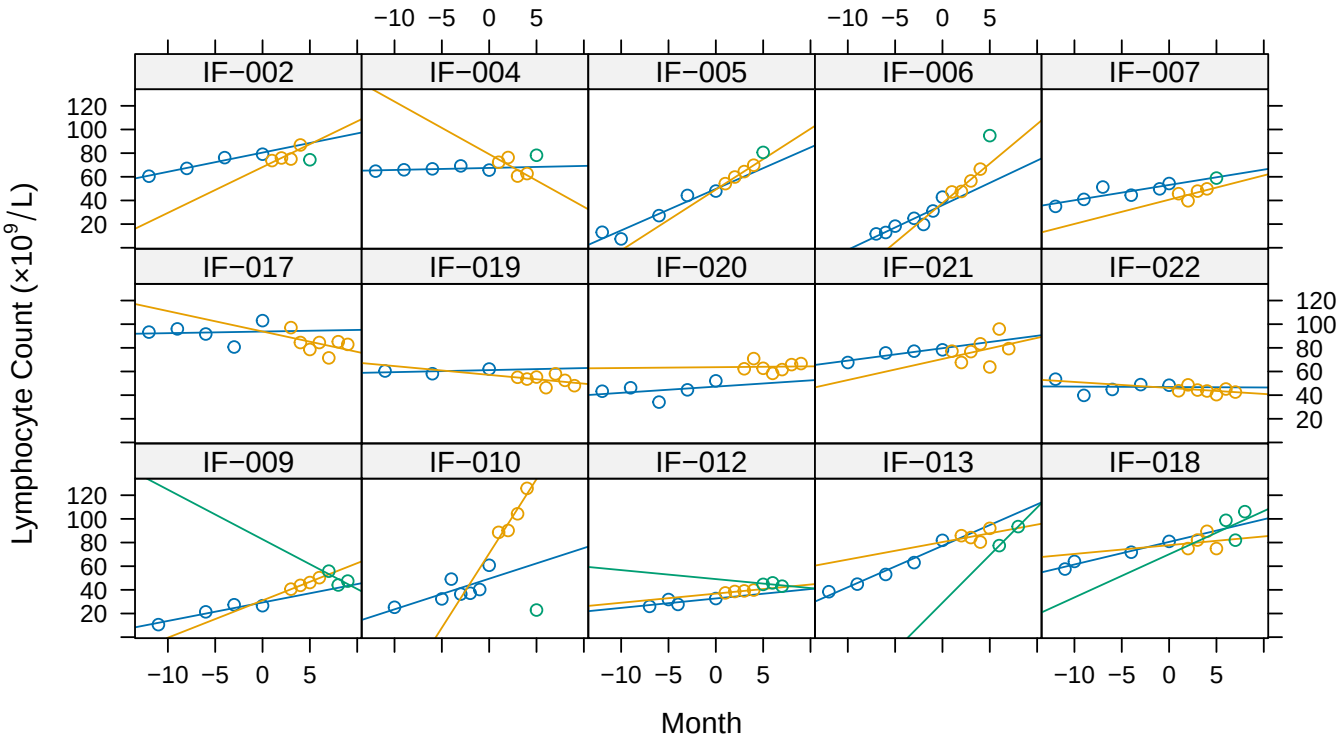
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1. Data description

The longitudinal data are from an experiment to study the effect of intermittent fasting (IF) on the lymphocyte measurements of 15 participants, each of whom was repeatedly measured before, during, and after the intervention. The data set (long format) consists of six columns:

- **LYMP** - [numeric] lymphocyte count ($\times 10^9$/L); lower values usually correspond to better outcomes;
- **ATG** - [numeric] relative net LC3-II expression; higher levels are associated with increased autophagy;
- **ID** - [factor] participant identifier; 15 levels;
- **Time** - [numeric] centered time point, with a frequency of 30 days, treating the **last** pretest as Month 0;
- **Time_Within_Phase** - [numeric] time point within each phase, starting from Month 0;
- **Phase** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest.

```
df <- read.csv("~/Downloads/IF_v2.csv", stringsAsFactors=T)
lattice::xyplot(LYMP ~ Time | ID, data=df, group=Phase, type=c("p","r"), layout=c(5,3),
                index.cond=list(c(6:9,11,10,12:15,1:5)), #change panel order
                xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
```



2. Linear mixed-effects regression (Part I)

The hypothesis is that IF can help patients lower their lymphocyte count. The objectives are then to investigate whether (1) the mean responses and (2) the linear slopes of time are statistically different before versus during the intervention.

2.1 Using the "*nlme*" package (v 3.1-162)

```
dat <- df[!is.na(df$LYMP) & df$Phase!=2,] #IF-013 contains one missing value

fit1 <- nlme::lme(fixed=LYMP ~ factor(Phase),
                  random=~1 | ID,         #random intercept for each subject
                  data=dat, method="ML")
summary(fit1)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1224.717 1236.706 -608.3586
##
## Random effects:
##  Formula: ~1 | ID
##         (Intercept) Residual
## StdDev:    15.96643 12.81501
##
## Fixed effects:  LYMP ~ factor(Phase)
##                   Value Std.Error  DF   t-value p-value
## (Intercept)    50.17287  4.422792 132 11.344163       0
## factor(Phase)1 14.12315  2.161386 132  6.534303       0
##  Correlation:
##                (Intr)
## factor(Phase)1 -0.246
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -2.49165610 -0.55618518 -0.03192352  0.62292969  4.25175278
##
## Number of Observations: 148
## Number of Groups: 15
```

`~1 | ID` is the random effects part of the model, allowing for the possibility that different subjects have different baseline lymphocyte counts. `LYMP ~ factor(Phase)` specifies the fixed effect, expecting the dependent variable to be influenced by the categorical treatment identifier.

This tutorial model (`fit1`) compares the dependent variable (`LYMP`) between the different phases of the experiment (`factor(Phase)`, dummy variable), while controlling for the repeated measures within each subject.

Two methods are available to estimate the coefficients: the restricted maximum likelihood (REML) by default and the maximum likelihood (ML). In a nutshell, ML is generally preferred when comparing models with different fixed effects.

```r
fit2 <- nlme::lme(fixed=LYMP ~ factor(Phase) * Time,
                  random=~Time | ID,      #random intercept and slope for each subject
                  data=dat, method="ML")
summary(fit2)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##       AIC      BIC    logLik
##   1124.47 1148.447 -554.2349
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.522600 (Intr)
## Time         2.019187 -0.089
## Residual     7.148936
##
## Fixed effects:  LYMP ~ factor(Phase) * Time
##                       Value Std.Error  DF   t-value p-value
## (Intercept)         57.89561  4.558001 130 12.701974  0.0000
## factor(Phase)1       3.71169  2.519625 130  1.473110  0.1431
## Time                 1.49179  0.573226 130  2.602454  0.0103
## factor(Phase)1:Time -0.13552  0.579038 130 -0.234052  0.8153
##  Correlation:
##                     (Intr) fc(P)1 Time
## factor(Phase)1      -0.177
## Time                 0.019 -0.178
## factor(Phase)1:Time -0.100 -0.514 -0.144
##
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max
## -3.26424016 -0.35417230 -0.04107843  0.49755880  3.70258899
##
## Number of Observations: 148
## Number of Groups: 15
```

`~Time | ID` adds random slopes for the time variable, which will allow the effect of time to vary across subjects. The interaction term `factor(Phase) * Time` allows the slope of the time trend to differ between the phases.

```r
(coef.fixed <- coef(summary(fit2))[,"Value"])
```

```
##         (Intercept)      factor(Phase)1                Time factor(Phase)1:Time
##           57.8956070            3.7116857            1.4917932           -0.1355249
```

```r
fit2u <- nlme::lme(fixed=LYMP ~ factor(Phase) * Time, random=~factor(Phase) * Time | ID,
                   data=dat, method="ML",
                   control=nlme::lmeControl(opt="optim", optCtrl=list(maxit=1e5)))

fit3u <- nlme::lme(fixed=LYMP ~ factor(Phase) + Time, random=~factor(Phase) + Time | ID,
                   data=dat, method="ML")
```

Furthermore, `~factor(Phase) * Time | ID` specifies a model (`fit2u`) with an interaction term as a random effect. This specification would allow the intercept, the slope for time, the slope for phase, and the slope for the interaction between time and phase to vary across subjects. However, please note that this model is quite complex and may require a large amount of data to estimate reliably. The `boundary (singular) fit` warning indicates that the estimated variance-covariance matrix is less than full rank.

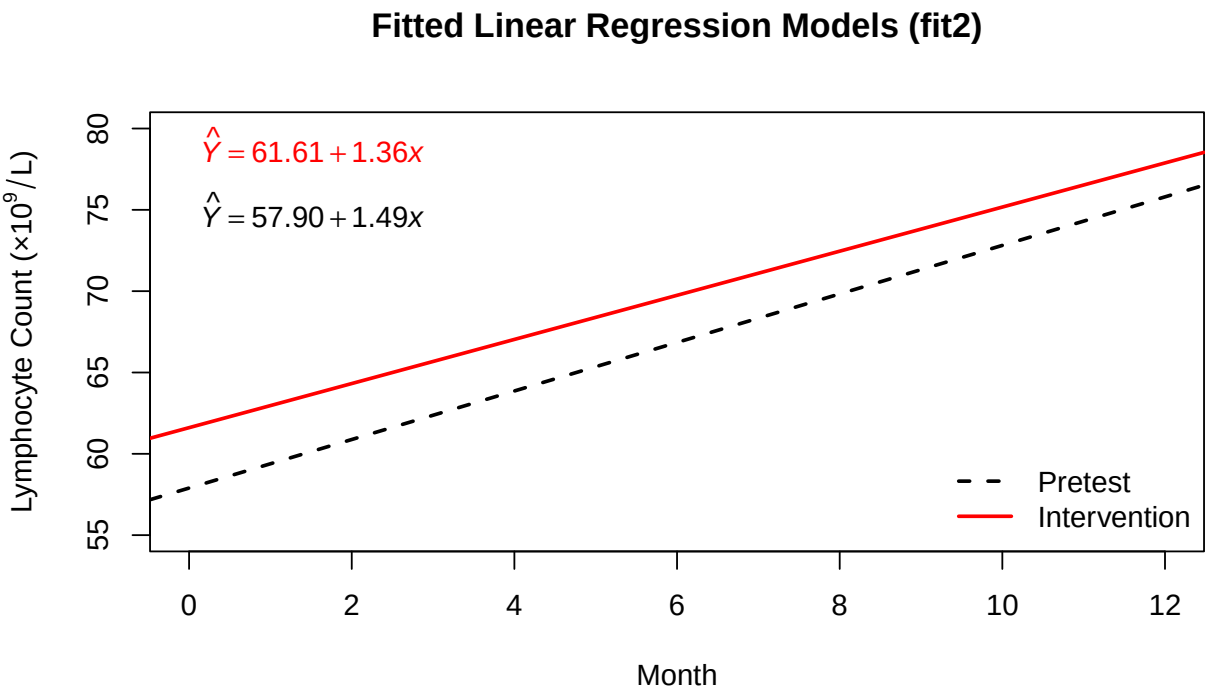
```
coef(fit2)
```

```
##        (Intercept) factor(Phase)1        Time factor(Phase)1:Time
## IF-002    73.87633       3.711686  0.93147491          -0.1355249
## IF-004    65.28458       3.711686 -0.09659144          -0.1355249
## IF-005    49.86903       3.711686  3.47626799          -0.1355249
## IF-006    39.00678       3.711686  4.29918136          -0.1355249
## IF-007    45.69031       3.711686  0.39434789          -0.1355249
## IF-009    32.89688       3.711686  2.02408234          -0.1355249
## IF-010    70.11684       3.711686  6.51099919          -0.1355249
## IF-012    33.75998       3.711686  0.98786257          -0.1355249
## IF-013    72.62868       3.711686  2.87679601          -0.1355249
## IF-017    86.11678       3.711686 -0.80831224          -0.1355249
## IF-018    74.08746       3.711686  1.16819162          -0.1355249
## IF-019    54.58306       3.711686 -0.63209714          -0.1355249
## IF-020    52.89076       3.711686  1.21778064          -0.1355249
## IF-021    73.98122       3.711686  0.43660057          -0.1355249
## IF-022    43.64540       3.711686 -0.40968697          -0.1355249
```

The function `coef` is extracting individual fitted intercept and slope in each phase. For example, IF-002 has a lymphocyte baseline of 73.87633 and a growth rate of 0.93147491 before the intervention; and a lymphocyte baseline of $73.87633 + 3.711686 = 77.58802$ and a growth rate of $0.93147491 - 0.1355249 = 0.79595$ during the intervention. IF-004, 017, 019, and 022 have a negative growth rate.

These values are a combination of the fixed and random effects. Use `coef` and `summary` to obtain the coefficients of the fixed effects, which are the mean fitted intercept and slope of all 15 subjects. In this case, the mean lymphocyte baseline is 57.8956070, and the mean growth rate is 1.4917932 before the intervention. The mean lymphocyte baseline becomes $57.8956070 + 3.7116857 = 61.60729$, and the mean growth rate becomes $1.4917932 - 0.1355249 = 1.356268$ during the intervention.

```
par(mar=c(5,5,5,5))
plot(1, type="n", xlim=c(0, 12), ylim=c(55, 80), main="Fitted Linear Regression Models (fit2)",
     xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
abline(a=coef.fixed[1], b=coef.fixed[3], lty=2, lwd=2)
abline(a=coef.fixed[1]+coef.fixed[2], b=coef.fixed[3]+coef.fixed[4], col="red", lwd=2)
legend("bottomright", c("Pretest", "Intervention"),
       col=c("black","red"), lty=c(2,1), lwd=2, bty="n")
text(1.5,75,bquote(hat(italic(Y)) ==
                   .(format(coef.fixed[1],digits=4,nsmall=2))
                   + .(round(coef.fixed[3],2)) * italic(x)))
text(1.5,79,bquote(hat(italic(Y)) == .(round(coef.fixed[1]+coef.fixed[2],2))
                 + .(round(coef.fixed[3]+coef.fixed[4],2)) * italic(x)), col="red")
```



2.2 Using the "*lme4*" package (v 1.1-33)

```
fit1 <- lme4::lmer(LYMP ~ factor(Phase) + (1 | ID), dat, REML=FALSE) #(same) output omitted

fit2 <- lme4::lmer(LYMP ~ factor(Phase) * Time + (Time | ID), dat, REML=FALSE)

fit3 <- lme4::lmer(LYMP ~ factor(Phase) + Time + (Time | ID), dat, REML=FALSE)

fit2u <- lme4::lmer(LYMP ~ factor(Phase) * Time + (factor(Phase) * Time | ID), dat, REML=F,
                    control=lme4::lmerControl(optCtrl=list(maxfun=1e5)))

fit3u <- lme4::lmer(LYMP ~ factor(Phase) + Time + (factor(Phase) + Time | ID), dat, REML=F)
```

In a linear mixed-effects regression (LMER), each subject has its intercept and slope (i.e., random intercepts and random slopes).

The p-value of .8153 for the interaction term in `fit2` is not statistically significant at an alpha level of .05 ($t = -0.234052$, $df = 130$). Hence, fail to reject that the linear slopes of time are the same before versus during the intervention.

▶ In summary, the mean lymphocyte count continued to increase at the same rate after participants received IF treatment.

If the slope of the time trend is the same across phases, one can remove the interaction term and specify `factor(Phase) + Time` as fixed effects.

```
fit3 <- nlme::lme(fixed=LYMP ~ factor(Phase) + Time,
                  random=~Time | ID,
                  data=dat, method="ML")
summary(fit3)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1122.526 1143.506 -554.2628
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 16.508068 (Intr)
## Time         2.026421 -0.087
## Residual     7.148179
##
## Fixed effects:  LYMP ~ factor(Phase) + Time
##                   Value Std.Error  DF   t-value p-value
## (Intercept)    57.79047  4.515995 131 12.796839  0.0000
## factor(Phase)1  3.40709  2.152902 131  1.582559  0.1159
## Time            1.47275  0.567068 131  2.597142  0.0105
##  Correlation:
##                (Intr) fc(P)1
## factor(Phase)1 -0.267
## Time            0.007 -0.296
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -3.2625501 -0.3584044 -0.0429246  0.5105937  3.6754491
##
## Number of Observations: 148
## Number of Groups: 15
```

3. Linear mixed-effects regression (Part II)

Investigate (3) any difference in either autophagy or lymphocyte count between those who fasted for three versus six months.

```
lymphocyte <- df[df$Phase==1, -2]
autophagy <- df[!is.na(df$ATG) & df$Phase==1, -1] #Excluded IF-007 and 012
fast6 <- paste0("IF-0",c(17,19,20,21,22))
lymphocyte$Fast <- factor(ifelse(lymphocyte$ID %in% fast6, "Six", "Three"))
autophagy$Fast <- factor(ifelse(autophagy$ID %in% fast6, "Six", "Three"))

fit.LYMP <- nlme::lme(fixed=LYMP ~ Fast * Time_Within_Phase,
                      random=~Time_Within_Phase | ID,
                      data=lymphocyte, method="ML")
summary(fit.LYMP)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: lymphocyte
##        AIC      BIC    logLik
##   564.0117 582.5516 -274.0059
```

```
##
## Random effects:
##  Formula: ~Time_Within_Phase | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##                    StdDev    Corr
## (Intercept)       16.516612 (Intr)
## Time_Within_Phase  1.528240 0.432
## Residual           6.191262
##
## Fixed effects:  LYMP ~ Fast * Time_Within_Phase
##                                Value Std.Error DF   t-value p-value
## (Intercept)                 65.07029  7.835393 58  8.304662  0.0000
## FastThree                   -3.84529  9.645958 13 -0.398642  0.6966
## Time_Within_Phase           -0.22457  0.884671 58 -0.253848  0.8005
## FastThree:Time_Within_Phase  3.42457  1.356162 58  2.525193  0.0143
##  Correlation:
##                             (Intr) FstThr Tm_W_P
## FastThree                   -0.812
## Time_Within_Phase            0.207 -0.168
## FastThree:Time_Within_Phase -0.135  0.105 -0.652
##
## Standardized Within-Group Residuals:
##          Min           Q1          Med           Q3          Max
## -2.417873860 -0.422707000  0.007624772  0.396155990  2.584876669
##
## Number of Observations: 75
## Number of Groups: 15
```

```r
fit.ATG <- nlme::lme(fixed=ATG ~ Fast * Time_Within_Phase,
                     random=~Time_Within_Phase | ID,
                     data=autophagy, method="ML")
summary(fit.ATG)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: autophagy
##         AIC      BIC   logLik
##   -5.651109 11.23588 10.82555
##
## Random effects:
##  Formula: ~Time_Within_Phase | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##                    StdDev     Corr
## (Intercept)       0.30856306 (Intr)
## Time_Within_Phase 0.05083683 -0.917
## Residual          0.15131001
##
## Fixed effects:  ATG ~ Fast * Time_Within_Phase
##                                  Value  Std.Error DF   t-value p-value
## (Intercept)                  0.6166054 0.15183125 46  4.061123  0.0002
## FastThree                    0.0722235 0.19562025 11  0.369203  0.7190
## Time_Within_Phase           -0.0312958 0.02767315 46 -1.130910  0.2640
## FastThree:Time_Within_Phase -0.0722687 0.04231507 46 -1.707871  0.0944
##  Correlation:
```

```
##                               (Intr) FstThr Tm_W_P
## FastThree                     -0.776
## Time_Within_Phase             -0.882  0.685
## FastThree:Time_Within_Phase    0.577 -0.808 -0.654
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -1.5396561 -0.5850891 -0.1483048  0.5270094  2.6615711
##
## Number of Observations: 61
## Number of Groups: 13
```

▶ The p-value of .0143 for the interaction term in `fit.LYMP` is statistically significant at an alpha level of .05 ($t = 2.525193$, $df = 58$). Hence, reject that the linear slopes of time are the same between the two fasting cohorts. Participants who fasted for three months showed a higher lymphocyte growth rate than those who fasted for six months during the intervention.

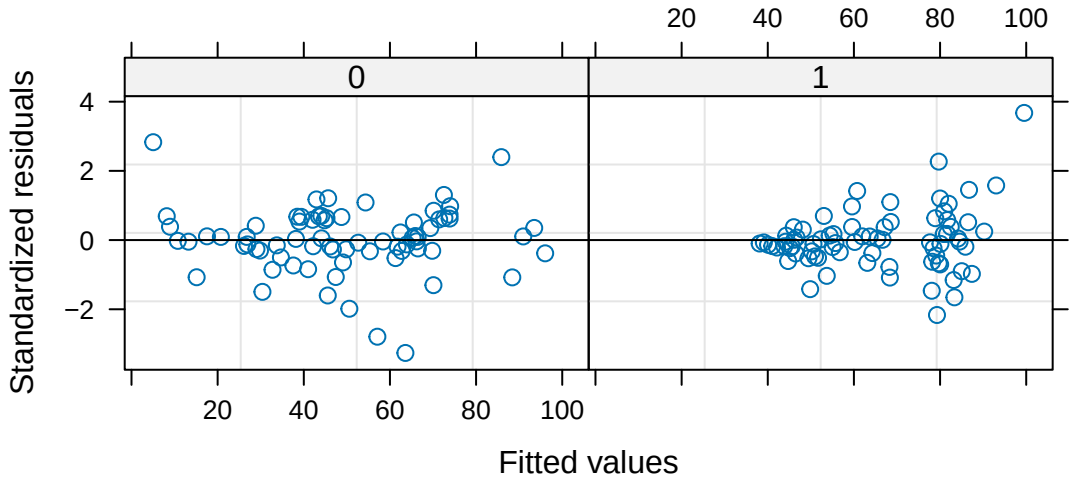
▶ The p-value of .0944 for the interaction term in `fit.ATG` is not statistically significant at an alpha level of .05 ($t = -1.707871$, $df = 46$). Hence, fail to reject that the linear slopes of time are the same between the two fasting cohorts. There was no difference between the autophagy values of fasting for three and six months.

4. Diagnostics

4.1 Check for linearity and homoscedasticity

Plot standardized residuals versus fitted values by phase. One should see a random scattering of points if the assumptions are met. Plus, check for influential outliers.

```
plot(fit3, resid(., type = "p") ~ fitted(.) | factor(Phase), abline=0)
```



The intervention phase (right panel, `Phase==1`) may witness heteroscedasticity according to the residual plot. Using the `weights` argument, one can incorporate a variance function structure into the model. The variance structure `varIdent(form=~1 | factor(Phase))` allows for different variances for each phase level.

```
fit4 <- nlme::lme(fixed=LYMP ~ factor(Phase) + Time,
                  random=~Time | ID,
                  cor=nlme::corCompSymm(),
                  weights=nlme::varIdent(form=~1 | factor(Phase)),
                  data=dat, method="ML")
summary(fit4)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: dat
##        AIC      BIC    logLik
##   1116.744 1143.719 -549.3719
##
## Random effects:
##  Formula: ~Time | ID
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr
## (Intercept) 18.654109 (Intr)
## Time         1.740448 0.701
## Residual    24.150594
##
## Correlation Structure: Compound symmetry
##  Formula: ~1 | ID
```

```
##  Parameter estimate(s):
##       Rho
## 0.8934861
## Variance function:
##  Structure: Different standard deviations per stratum
##  Formula: ~1 | factor(Phase)
##  Parameter estimates:
##         0         1
## 1.0000000 0.6887895
## Fixed effects:  LYMP ~ factor(Phase) + Time
##                   Value Std.Error  DF  t-value p-value
## (Intercept)    58.15180  7.825701 131 7.430874  0.0000
## factor(Phase)1  2.96753  2.848364 131 1.041837  0.2994
## Time            1.46597  0.500614 131 2.928351  0.0040
##  Correlation:
##                 (Intr) fc(P)1
## factor(Phase)1 -0.625
## Time            0.454 -0.269
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -2.6474784 -0.5986648  0.1343088  0.6488575  2.1497415
##
## Number of Observations: 148
## Number of Groups: 15
```

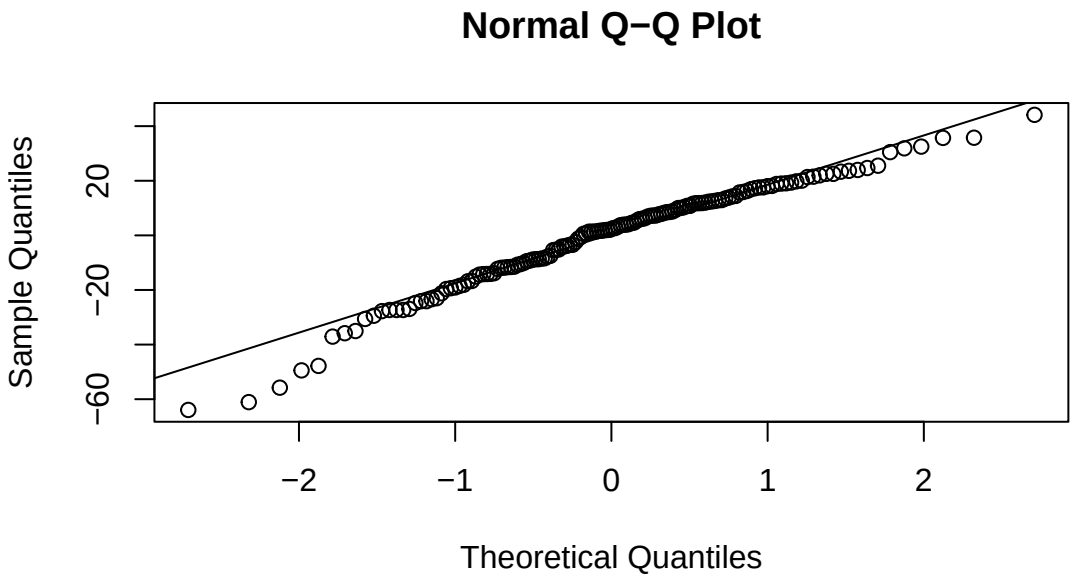
```
coef(fit4) #final model
```

```
##        (Intercept) factor(Phase)1       Time
## IF-002    59.32070        2.96753  1.4111258
## IF-004    55.62287        2.96753  0.2255899
## IF-005    62.31410        2.96753  3.0766349
## IF-006    62.14011        2.96753  3.3017775
## IF-007    38.40942        2.96753  0.3976132
## IF-009    49.97750        2.96753  1.4889376
## IF-010   110.22773        2.96753  6.0195517
## IF-012    43.41974        2.96753  0.4736706
## IF-013    69.84448        2.96753  2.9312904
## IF-017    58.85621        2.96753 -0.1613294
## IF-018    61.64983        2.96753  1.5173742
## IF-019    44.07036        2.96753 -0.4114820
## IF-020    59.57141        2.96753  1.0633564
## IF-021    56.31475        2.96753  1.0211253
## IF-022    40.53781        2.96753 -0.3656464
```

4.2 Check for normality of the residuals

The residuals should follow a straight line in a QQ plot (Quantile-Quantile plot) if normally distributed.

```
qqnorm(resid(fit4)); qqline(resid(fit4)) # shapiro.test(resid(fit4))
```



4.3 Check for random effects assumptions

Check that the random effects are zero-centered and normally distributed, and have a constant variance. Use `ranef` to extract the random effects from the model and plot histograms.

```
rand.ef <- nlme::ranef(fit4)$Time #15 subjects
mean(rand.ef)
```

```
## [1] 1.657933e-15
```

```
hist(rand.ef, prob=TRUE, col="white", yaxt="n",
     ylab="", xlab="Subject-Specific Random Effects (15)", main="")
lines(density(rand.ef),col="red",lwd=2)
```

